INTRA-AMNIOTIC INFECTION

Dr. Hussen T (MD, OBGYN Resident)

March 2024

ETIOPATHOGENESIS

ASCENDING INFECTION

- More commonly caused by organisms that are part of the normal vaginal flora.
    1. GBS
    2. E. coli,
    3. Bacteroides
    4. Prevotella species
    5. Ureaplasma urealyticum.

HEMATOGENOUS DISSEMINATION

- Less common
    - Listeria monocytogens

EPIDEMIOLOGY

- IAI occurs in:
  - **25%** preterm deliveries.
  - **1-5%** of term deliveries.

RISK FACTORS

- Low socioeconomic status,
- Young age
- Nulliparity
- Ruptured membranes,
- Multiple vaginal examinations, and
- Pre-existing infections of the lower genital tract(bacterial vaginosis & GBS infection)
- Urogenital tract infections,
- Extended duration of labor
- Poor maternal nutrition with a low BMI(<19.8 kg/m2), and
- Cigarette smoking.
- Obstetric procedures, such as a Cerclage, amniocentesis, and percutaneous umbilical blood sampling, which introduce microorganisms from the skin or vaginal flora into the uterine cavity.

CLINICAL FINDINGS

MATERNAL	FETAL
• Maternal fever • Maternal tachycardia • Purulent Cx discharge	• Fetal tachycardia

DIAGNOSIS OF IAI

- **In term patients** who are progressing to delivery, laboratory confirmation of the dx of IAI is <u>not routinely necessary</u>
- **In preterm patients** who are being evaluated for tocolysis or ACS, when IAI is suspected, laboratory assessment may be of value in establishing the diagnosis.
  - In this clinical context, amniotic fluid should be obtained by **transabdominal amniocentesis.**

ACOG-2024

- Clinical guidance generally has incorporated maternal fever as a required criterion for suspected Intraamniotic infection.
- The diagnosis of **isolated maternal fever** is
  - T^0: 38.0°C—38.9°C **with no** additional risk factors

    →With or without persistent temperature elevation

- The diagnosis of **suspected IAI** currently include:
  - T^0 ≥39.0°C **or**
  - T^0:38.0–38.9°C **with** 1-additional clinical risk factor.

Features of **isolated maternal fever** and **triple I** with classification

Isolated maternal fever ("documented" fever)	• T^0 ≥38.0°C (2-occasions 30' apart)
Suspected triple I	• T^0 ≥39.0°C alone **OR** • T^0:38.0–38.9°C + any of the ff: 1. Purulent Cx discharge 2. Fetal tachycardia **3. Leucocytosis(15k)**
Confirmed triple I	All of the above **+** objective **lab findings** of infection, such as: **1.** +ve AF **Gram stain** for bacteria **2.** +ve AF **Culture** results **3.** ↑**AF WBC** (>**30** cells/mm^3) in the absence of a bloody tap **4.** ↓AF **glucose** (**≤14** mg/dL) or 5. Histopathologic evidence of infection/inflammation or both • In the placenta, fetal membranes, or cord vessels (funisitis)

TABLE 58.5 Diagnostic Tests for Chorioamnionitis

Test	Abnormal Finding	Comment
Maternal WBC	>15,000 cells/mm^3 with a preponderance of neutrophils	Labor and/or steroids may cause an elevation of the WBC
Amniotic fluid glucose	≤10–15 mg/dL	*Excellent* correlation with positive amniotic fluid culture and clinical infection
Amniotic fluid IL-6	≥7.9 ng/mL	Same as above
Amniotic fluid leukocyte esterase	≥1+ reaction	*Good* correlation with amniotic fluid culture and clinical infection
Amniotic fluid Gram stain	Any organism in an oil immersion field	Test is very sensitive to inoculum effect; it cannot identify mycoplasmas
Amniotic fluid culture	Growth of any aerobic or anaerobic pathogen	Gold standard for confirmation of bacterial infection, but not immediately available

COMPLICATIONS OF IAI

Maternal Complications

- **Bacteremia** occurs in 3-12%
- Uterine atony,
- Dysfunctional labor
- Septic pelvic vein thrombophlebitis, and
- Pelvic abscess
  When cesarean delivery is required,
  - **8%** develop a wound infection
  - **1%** develop a pelvic abscess

Fetal & Neonatal Complications

- EONS: 5 - 10%
- Meningitis: 1%
- Neonatal seizures

MANAGEMENT

- The most extensively tested IV antibiotic regimen for the treatment of chorioamnionitis is the combination of:
  - **Ampicillin** (2 g QID) or penicillin (5 million U QID) plus
  - **Gentamicin** (1.5 mg/kg TID)

- This specifically target the 2-organisms most likely to cause neonatal infection: **GBS & E. coli**.
- In patients who are allergic to β-lactam antibiotics, clindamycin (900 mg every 8 hours) should be substituted for ampicillin.
- If a woman with chorioamnionitis requires **cesarean delivery**, a drug with activity against anaerobic organisms should be added to the antibiotic regimen.
  - Either **Clindamycin (900 mg)** or **Metronidazole (500 mg**) is an excellent choice for this purpose.
- Failure to provide effective anaerobic coverage may result in treatment failures in 20% to 30% of patients

TABLE 48.4 Recommended Antibiotic Coverage for Suspected Intraamniotic Infection

Recommended Antibiotic	Dosage
No penicillin allergy[a]	
• Ampicillin *plus*	• 2 g IV q 6 hours
• Gentamicin	• 2 mg/kg IV load → 1.5 mg/kg q 8 hours OR 5 mg/kg IV q 24 hours
Mild penicillin allergy[a]	
• Cefazoline *plus*	• 2 g IV q 8 hours
• Gentamicin	• 2 mg/kg IV load → 1.5 mg/kg q 8 hours OR 5 mg/kg IV q 24 hours
Severe penicillin allergy[a]	
• Clindamycin *or*	• 900 mg IV q 8 hours
• Vancomycin *plus*	• 1 g IV q 12 hours
• Gentamicin	• 2 mg/kg IV load → 1.5 mg/kg q 8 hours OR • 5 mg/kg IV q 24 hours
Alternative regimens[b]	
• Ampicillin-sulbactam	• 3 g IV q 6 hours
• Piperacillin-tazobactam	• 3.375 g IV q 6 hours or 4.5 g IV q 8 hours
• Cefotetan	• 2 g IV q 12 hours
• Cefoxitin	• 2 g IV every 8 hours
• Ertapenem	• 1 g IV every 24 hours

PERINATAL OUTCOME

- Chorioamnionitis is associated with increased risks of :
  - RDS,
  - Periventricular leukomalacia, and
  - Cerebral palsy.

Treatment of chorioamnionitis:

- Give a combination of antibiotics until the woman gives birth. **Note** that metronidazole should be added if the route of delivery is CS to cover anaerobic organisms.
- *Option 1:* Ampicillin 2 g IV every six hours PLUS gentamycin 5 mg/kg body weight IV every 24 hours ± metronidazole 500 mg IV TID
- *Option 2:* Ceftriaxone 1 gm IV BID for 10 days ± metronidazole 500 mg IV TID
- After delivery, shift the antibiotics to PO medication after the symptoms and signs of infection have subsided for 48 hours.
- Emergency priming and induction of labor if there is no contraindication for vaginal delivery.
- If conditions for vaginal delivery are not met, perform C/S. (Before procedure, cleanse the vagina with povidone-iodine)
- Newborn requires evaluation and management in NICU.

PREVENTION

- GBS screening
- UTI treatment

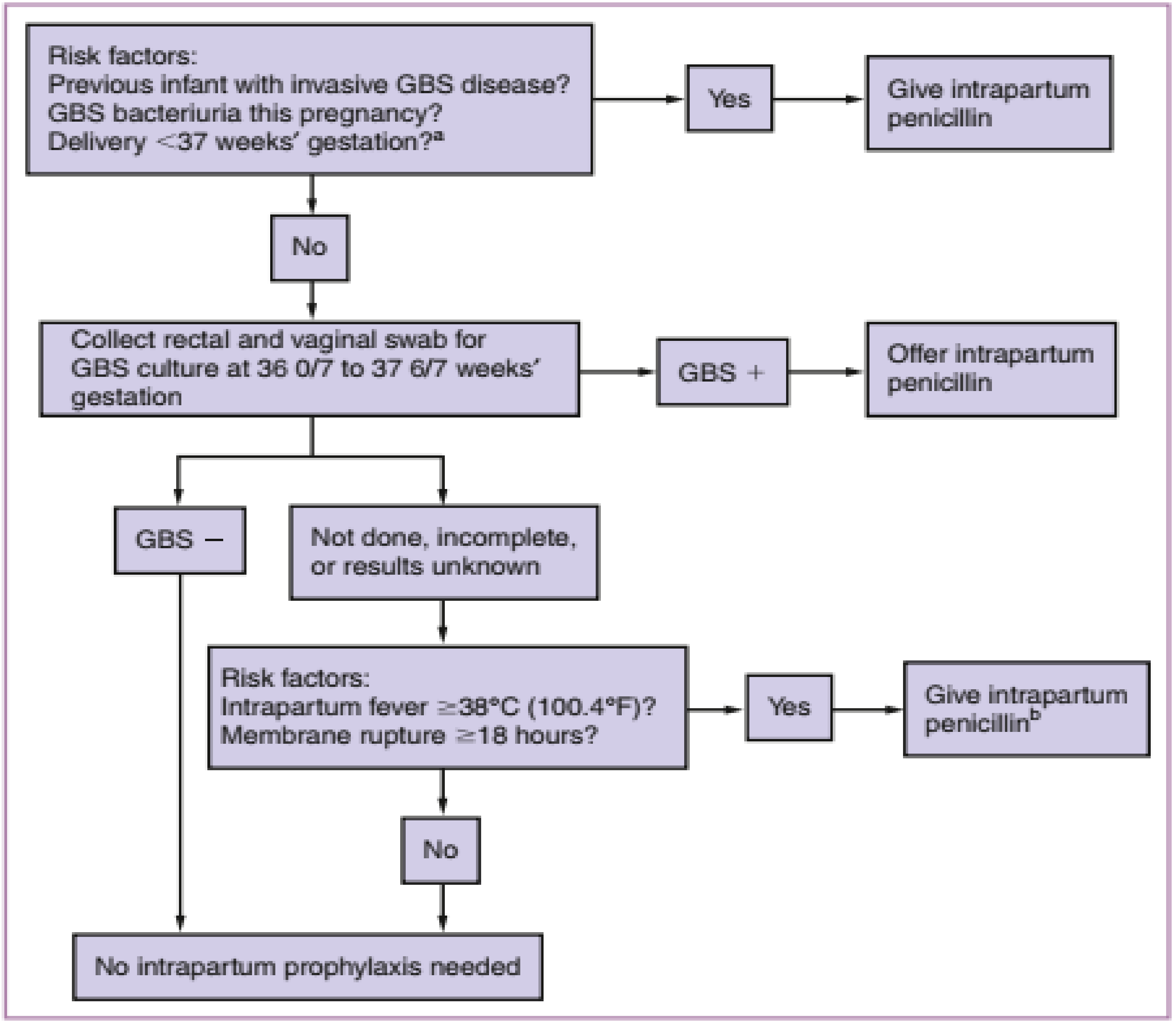

Figure 48.3 Algorithm for prevention of early-onset group B streptococcal (GBS) disease in neonates with prenatal screening at 36 0/7 to 37 6/7 weeks' gestation. [a]If membranes rupture before 37 weeks' gestation and the mother has not begun labor, collect GBS culture and administer antibiotics until cultures are completed and the results are negative. [b]Broader-spectrum antibiotics may be considered at the physician's discretion, based on clinical indications (e.g., ampicillin plus gentamicin to treat clinical chorioamnionitis). (*Modified from Centers for Disease Control and Prevention. Prevention of perinatal group B streptococcal disease: a public health perspective.* MMWR Recomm Rep. *1996;45[RR-7]:1.*)

- PROM complicates about **8-10%** of pregnancies and is a significant cause of gestational age–dependent and infectious perinatal morbidity and mortality.
- **Latency** from ROM to delivery is typically **brief** and ↓ with ↑GA at membrane rupture.
- **IAI** is common after preterm PROM and ↑in frequency with ↓GA at membrane rupture.
- Women with a prior preterm birth due to PROM have:
  - **3.3-fold** higher risk for subsequent preterm birth due to PROM and
  - **13.5-fold** higher risk for PROM before 28 weeks' gestation.
- **Vaginally collected AF** can reliably predict the presence of fetal pulmonary maturity after preterm PROM.
  - **Conservative management after PROM with documented fetal pulmonary maturity near term (32-36wks**) prolongs latency only briefly but ↑ the risk for perinatal infection and **does not improve newborn outcomes**.
- ACS administration and broad-spectrum antibiotic administration have been shown to ↓ newborn complications.
- **Cervical cerclage retention** after preterm PROM has not been shown to improve newborn outcomes.
- **Lethal pulmonary hypoplasia** is common after PROM that occurs before 20 weeks and can be predicted with serial ultrasound assessment of lung and chest growth.

SUMMARY

- PROM complicates about **8-10%** of pregnancies and is a significant cause of gestational age–dependent and infectious perinatal morbidity and mortality.
- **Latency** from ROM to delivery is typically **brief** and ↓ with ↑GA at membrane rupture.
- **IAI** is common after preterm PROM and ↑in frequency with ↓GA at membrane rupture.
- Women with a prior preterm birth due to PROM have:
  - **3.3-fold** higher risk for subsequent preterm birth due to PROM and
  - **13.5-fold** higher risk for PROM before 28 weeks' gestation.
- **Vaginally collected AF** can reliably predict the presence of fetal pulmonary maturity after preterm PROM.
  - **Conservative management after PROM with documented fetal pulmonary maturity near term (32-36wks**) prolongs latency only briefly but ↑ the risk for perinatal infection and **does not improve newborn outcomes**.
- ACS administration and broad-spectrum antibiotic administration have been shown to ↓ newborn complications.
- **Cervical cerclage retention** after preterm PROM has not been shown to improve newborn outcomes.
- **Lethal pulmonary hypoplasia** is common after PROM that occurs before 20 weeks and can be predicted with serial ultrasound assessment of lung and chest growth.

REFERENCES

1. American College of Obstetricians and Gynecologists. ACOG Committee practice bulletin number 831: Committee opinion summary. Obstet Gynecol. February 2021; 138: p166-167.
2. Society for Maternal-Fetal Medicine (SMFM), Dashe JS1, Pressman EK1, Hibbard JU1. SMFM Consult Series #46: Evaluation and management of polyhydramnios. Am J Obstet Gynecol. 2018 Oct;219(4).
3. F. Gray Cunningham, KJ Leveno, JS Dashe, BL Hoffman, CY Spong, BM Casey. *Williams* OBSTETRICS. 26th ed. New York; McGraw Hill, 2022. p 643-647.
4. Mark B. Landon, et al. Gabbe's obstetrics: normal and problem pregnancies. 8th ed. Philadelphia; Elseivar, 2021. p 539–544.
5. Khan S, Donnelly J. Outcome of pregnancy in women diagnosed with idiopathic polyhydramnios. Aust N Z J Obstet Gynaecol 2017;57:57–62.
6. Pilliod, Rachel A et al. “The risk of fetal death in nonanomalous pregnancies affected by polyhydramnios.” *American journal of obstetrics and gynecology* vol. 213,3 (2015): 410.e1-6.
7. Odibo IN, Newville TM, Ounpraseuth ST, et al. Idiopathic polyhydramnios: persistence across gestation and impact on pregnancy outcomes. Eur J Obstet Gynecol Reprod Biol 2016;199:175–8.
8. Luo Q-Q, Zou L, Gao H, Zheng Y-F, Zhao Y, Zhang W-Y. Idiopathic polyhydramnios at term and pregnancy outcomes: a multicenter observational study. J Matern Fetal Neonatal Med 2017;30:1755–9.
9. American College of Obstetricians and Gynecologists. Ultrasound in pregnancy. Practice bulletin no. 175. Obstet Gynecol 2016;128: e241–56.
10. Hamza A, Herr D, Solomayer EF, Meyberg-Solomayer G. Polyhydramnios: causes, diagnosis and therapy. Geburtshilfe Frauenheilkd 2013;73:1241–6.
11. Pilliod RA, Page JM, Burwick RM, Kaimal AJ, Cheng YW, Caughey AB. The risk of fetal death in nonanomalous pregnancies affected by polyhydramnios. Am J Obstet Gynecol 2015;213:410.e1–6.
12. Magann EF, Chauhan SP, Doherty D, Lutgendorf MA, Magann MI, Morrison JC. A review of idiopathic hydramnios and pregnancy outcomes. Obstet Gynecol Surv 2007;62:795–802.

THANK YOU